



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2013

Year 11 Mathematics
Yearly

Mathematics Extension Continuers

General Instruction

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Start each **NEW** question in a separate answer booklet.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 67

SECTION I – 7 Marks

- Attempt Questions 1 – 7

SECTION II – 60 Marks

- Attempt Questions 8 – 12

Examiner: *J. Chen*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SECTION I – 7 Marks

Attempt Questions 1 – 7

Answer this section on the Multiple Choice Answer Sheet

1)

$$\int (x^{-1/2} + 4). dx =$$

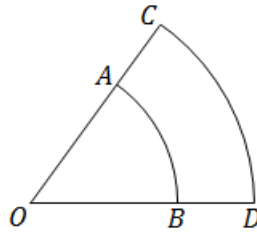
(A) $-\frac{3}{2}x^{-3/2} + 4x + C$

(B) $\frac{1}{2}\sqrt{x} + 4x + C$

(C) $2\sqrt{x} + 4x + C$

(D) $\frac{2}{3}x^{-3/2} + 4x + C$

- 2) In the figure, OAB and OCD are sectors with centre O . If the area of sector $OAB = 6\pi \text{ cm}^2$, the area of $ABCD = \frac{14}{3}\pi \text{ cm}^2$ and $OA = 6 \text{ cm}$, then $AC =$



(A) 2 cm

(B) 4 cm

(C) 6 cm

(D) 8 cm

3) $\frac{\sin 120^\circ}{1 + \sin(90^\circ + \theta)} - \frac{\sin 240^\circ}{1 - \cos(-\theta)} =$

(A) $\frac{1}{\sin^2 \theta}$

(B) $\frac{\sqrt{3}}{\sin^2 \theta}$

(C) $\frac{\sqrt{3}}{1 + \cos \theta}$

(D) $\frac{\sqrt{3}}{2(1 + \sin \theta)(1 + \cos \theta)}$

4) If $A = \frac{16}{r^2} - \pi\sqrt{r}$, then $\frac{dA}{dr} =$

(A) $\frac{8}{r} - \frac{\pi}{2\sqrt{r}}$

(B) $-\frac{8}{r} - \frac{\pi}{2\sqrt{r}}$

(C) $-\frac{32}{r^3} - \frac{\pi}{2\sqrt{r}}$

(D) $\frac{32}{r^3} + \frac{\pi}{2\sqrt{r}}$

5) If $2 \ln x = 6$, then x equals

(A) e^3

(B) $\sqrt{e^3}$

(C) e^{36}

(D) $2e^6$

6)

$$\int \frac{2}{3x+1} \cdot dx =$$

(A) $2 \log_e(3x+1) + C$

(B) $\frac{2}{3} \log_e(3x+1) + C$

(C) $6 \log_e(3x+1) + C$

(D) $\frac{3}{2} \log_e(3x+1) + C$

7) $\frac{d}{dx}[\cos(3x+1)] =$

(A) $\sin(3x+1)$

(B) $-\sin(3x+1)$

(C) $3 \sin(3x+1)$

(D) $-3 \sin(3x+1)$

End of SECTION I

SECTION II – 60 Marks

Attempt Questions 8 – 12

Answer each question in a SEPARATE writing booklet.

QUESTION 8 (12 marks) Use a SEPARATE writing booklet.

(a) Find the exact value of $\tan 75^\circ$. 2

(b) Solve the equation $2 \cos \theta = \sqrt{3}$, where $0 \leq \theta \leq 2\pi$. 2

(c) Let $y = x + \sin kx$. It is given that $\frac{dy}{dx} = 3$, when $x = 0$.

(i) Find the value of k . 2

(ii) Find $\frac{d^2y}{dx^2}$ 2

(d) Solve 2

$$\frac{1}{x+1} \geq 1$$

(e) Evaluate 2

$$\lim_{h \rightarrow 2} \frac{2h-4}{h^2-4}$$

End of Question 8

QUESTION 9 (12 marks) Use a SEPARATE writing booklet.

- (a) Prove that 2

$$2 \sin A \sin B = \cos(A - B) - \cos(A + B)$$

- (b) The region bounded by the curve $y = 2 + e^{-x}$, the x -axis, the y -axis and the line $x = 1$ is rotated about the x -axis. Find the volume of the solid generated. 2

- (c) Evaluate the following integrals.

(i)

$$\int_{-1}^0 \frac{1}{\sqrt{1-2x}} \cdot dx \quad \quad \quad 2$$

(ii)

$$\int_2^4 \frac{x}{x-1} dx \quad \quad \quad 2$$

- (d) The value of a car \$ C can be represented by the formula

$$C = 1000 + Ae^{-kt}$$

where t is the age in years from new.

- (i) The price of a brand new car is \$18000 and after 3 years, the same car has depreciated to \$7500. Find the value of A and k . 2

- (ii) Find the age of the car when its value first falls below \$3000. 2

End of Question 9

QUESTION 10 (12 marks) Use a SEPARATE writing booklet.

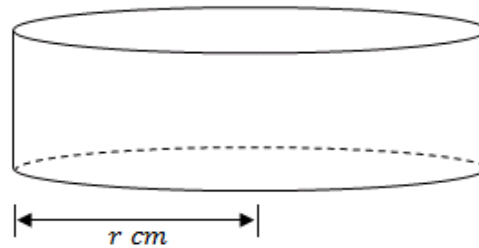
(a) Let $y = x^2 e^{-x}$, where $x > 0$. Show that $x \frac{d^2 y}{dx^2} - \frac{dy}{dx} - (x - 3)y = 0$. 3

(b) (i) Show that $\sin^2 x \cos^2 x = \frac{1 - \cos 4x}{8}$. 3

(ii) Hence find 2

$$\int \sin^2 x \cos^2 x \cdot dx$$

(c) The figure below shows a right circular cylindrical can without cover and its volume is $27\pi \text{ cm}^3$. Let $A \text{ cm}^2$ be the outer surface area of the circular cylindrical can and $r \text{ cm}$ be the radius of the can.



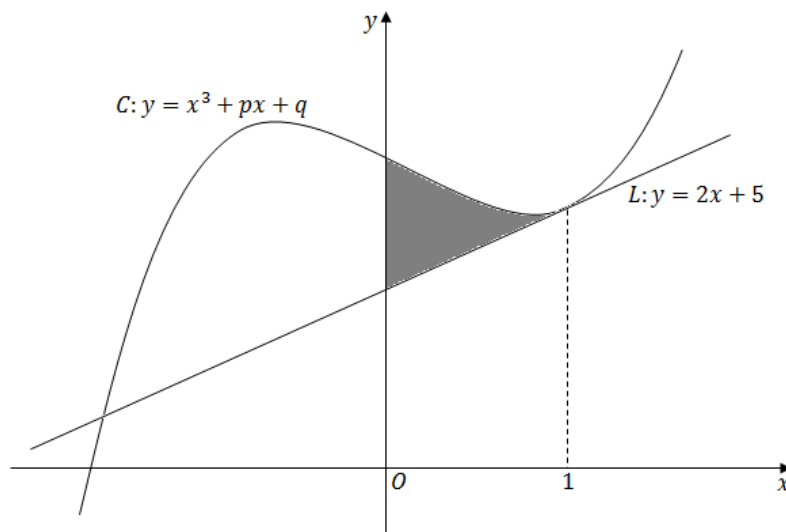
(i) Show that $A = \pi \left(\frac{54}{r} + r^2 \right)$. 2

(ii) Find r if A is minimum. 2

End of Question 10

QUESTION 11 (12 marks) Use a SEPARATE writing booklet.

- (a) The diagram below shows the graphs of the curve $C: y = x^3 + px + q$ and the line $L: y = 2x + 5$. L touches C at $x = 1$.



- (i) Find the values of p and q . 4
- (ii) Find the area of the shaded region. 2
- (b) Find $\frac{dy}{dx}$ of the following functions.
- (i) $y = \frac{x-2}{\sqrt{x+1}}$, where $x > -1$. 3
- (ii) $y = \ln[\ln(2x + 1)]$, where $x > 0$. 3

End of Question 11

QUESTION 12 (12 marks) Use a SEPARATE writing booklet.

(a) Let

3

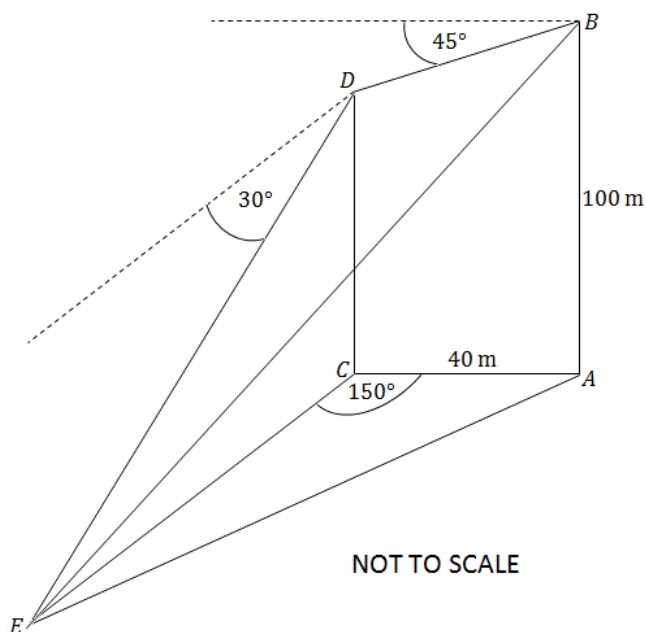
$$\int_a^b \frac{f(x)}{f(x) - g(x)} dx = 5.$$

Find the value of

$$\int_a^b \frac{g(x)}{f(x) - g(x)} dx$$

in terms of a and b .

- (b) The figure below shows two vertical towers AB and CD . The height of tower AB is 100m. The tower CD is 40m away from AB . The angle of depression of D from B is 45° . A canopy BDE is constructed to cover the two towers. A , C and E are on the horizontal ground such that $\angle ACE = 150^\circ$. The angle of depression of E from D is 30° .



- (i) Find the angle of elevation of B from E , correct your answer to nearest degree. 5
- (ii) Find the area of canopy BDE , correct your answer to 3 significant figures. 4

End of Question 12
End of Exam

Ext Continuers 11 Yearly 13

1) C

2) D

3) B

4) C

5) A

6) B

7) D

Q11 Actually Q8 solutions

$$(a) \tan 75^\circ = \frac{\tan 45^\circ + \tan 30^\circ}{1 - \tan 45^\circ \tan 30^\circ}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1}$$

$$= \frac{(\sqrt{3} + 1)^2}{2}$$

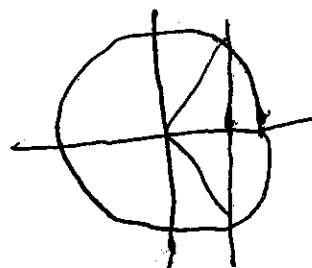
$$= \frac{4 + 2\sqrt{3}}{2}$$

$$= 2 + \sqrt{3}$$

[2]

$$(b) 2\cos\theta = \sqrt{3}$$

$$\cos\theta = \frac{\sqrt{3}}{2}$$



$$\theta = \frac{\pi}{6}, \frac{11\pi}{6} [2]$$

$$(c) y = x + \sin kx$$

$$\frac{dy}{dx} = 1 + k\cos kx$$

(i) When $x=0$

$$\frac{dy}{dx} = 3 = 1 + k \cdot 1$$

$$\therefore k = 2 [2]$$

$$(ii) \frac{dy}{dx} = 1 + 2\cos 2x$$

$$\frac{d^2y}{dx^2} = -4\sin 2x [2]$$

$$(d) \frac{1}{x+1} \gg 1, x \neq -1$$

Multiply by $(x+1)^2$

$$x+1 \gg (x+1)^2$$

$$x+1 \gg x^2 + 2x + 1$$

$$0 \gg x^2 + x$$

$$0 \gg x(x+1)$$

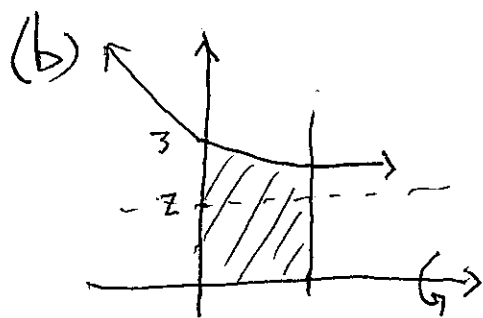
$$\therefore -1 < x \leq 0 [2]$$

Q11 (Cont'd)

$$\begin{aligned} (a) \lim_{h \rightarrow 2} \frac{2h-4}{h^2-4} &= \lim_{h \rightarrow 2} \frac{2(h-2)}{(h+2)(h-2)} \\ &= \lim_{h \rightarrow 2} \frac{2}{h+2} \\ &= \frac{1}{2} \quad [2] \end{aligned}$$

Q12 Actually Q9 solutions

$$\begin{aligned} (a) \cos(A-B) &= \cos A \cos B + \sin A \sin B \\ \cos(A+B) &= \cos A \cos B - \sin A \sin B \\ \therefore \cos(A-B) - \cos(A+B) &= 2 \sin A \sin B \quad [2] \end{aligned}$$



$$\begin{aligned} V &= \pi \int_0^1 (2+e^{-x})^2 dx \\ &= \pi \int_0^1 (4 + 4e^{-x} + e^{-2x}) dx \\ &= \pi \left[4x - 4e^{-x} - \frac{1}{2}e^{-2x} \right]_0^1 \\ &= \pi \left[\left(4 - \frac{4}{e} - \frac{1}{2e^2} \right) - \left(0 - 4 - \frac{1}{2} \right) \right] \\ &= \pi \left(8\frac{1}{2} - \frac{4}{e} - \frac{1}{2e^2} \right) \\ &\approx 21.868 \quad [2] \end{aligned}$$

(c)

$$\begin{aligned} (i) \int_{-1}^0 \frac{1}{\sqrt{1-2x}} dx &= \int_{-1}^0 (1-2x)^{-1/2} dx \\ &= \left[-\frac{1}{2} (1-2x)^{1/2} \right]_{-1}^0 \\ &= \left[-\sqrt{1-2x} \right]_{-1}^0 \\ &= (-\sqrt{1-0}) - (-\sqrt{1+2}) \\ &= \sqrt{3} - 1 \quad [2] \end{aligned}$$

$$\begin{aligned} (ii) \int_2^4 \frac{x}{x-1} dx &= \int_2^4 \frac{x-1+1}{x-1} dx \\ &= \int_2^4 \left(1 + \frac{1}{x-1} \right) dx \\ &= \left[x + \ln(x-1) \right]_2^4 \\ &= (4 + \ln 3) - (2 + \ln 1) \\ &= 2 + \ln 3 \quad [2] \end{aligned}$$

$$\begin{aligned} (d) C &= 1000 + A e^{-kt} \\ (i) \text{ When } t=0, C &= 18000 \\ \therefore 18000 &= 1000 + A.1 \\ A &= 17000 \end{aligned}$$

Q12 (Contd)

When $t=3$, $C=7500$

$$\therefore 7500 = 1000 + 17000e^{-3k}$$

$$e^{-3k} = \frac{6500}{17000}$$

$$-3k = \ln\left(\frac{6500}{17000}\right)$$

$$k = \frac{\ln\left(\frac{65}{170}\right)}{-3}$$

$$\therefore k = 0.32047... [2]$$

$$(ii) 3000 > 1000 + 17000e^{-kt}$$

$$\frac{2000}{17000} > e^{-kt}$$

$$-kt < \ln \frac{2}{17}$$

$$t > \frac{\ln \frac{2}{17}}{-k}$$

$$> 6.677...$$

$\therefore$ 6 yrs 9 months

or 7 yrs to nearest year.

[2]

Q13

$$(a) \quad y = x^2 e^{-x}, \quad x > 0.$$

$$\frac{dy}{dx} = 2x e^{-x} - x^2 e^{-x}$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= -2x e^{-x} + 2e^{-x} + x^2 e^{-x} - 2x e^{-x} \\ &= e^{-x} (x^2 - 4x + 2) \end{aligned}$$

$$LHS = x \frac{d^2 y}{dx^2} - \frac{dy}{dx} - (x-3)y$$

$$= x e^{-x} (x^2 - 4x + 2) - x e^{-x} (2 - x) - (x-3) x^2 e^{-x}$$

$$= x e^{-x} [x^2 - 4x + 2 - 2 + x - x^2 + 3x]$$

$$= x e^{-x} [x^2 - x^2 - 4x + 4x + 2 - 2]$$

$$= x e^{-x} \times 0$$

$$= 0$$

$$= RHS.$$

Q13(b) (b)

$$\begin{aligned} \text{(b)} \quad (i). \quad R.H.S. &= \frac{1 - \cos 4x}{8} \\ &= \frac{1 - (1 - 2\sin^2 2x)}{8} \\ &= \frac{2\sin^2 2x}{8} \\ &= \frac{\sin^2 2x}{4} \\ &= \frac{(2\sin x \cos x)^2}{4} \\ &= \sin^2 x \cos^2 x. \\ &= L.H.S. \end{aligned}$$

$$\begin{aligned} (ii) \quad \int \sin^2 x \cos^2 x \, dx &= \int \frac{(1 - \cos 4x)}{8} \, dx \\ &= \frac{1}{8} \int (1 - \cos 4x) \, dx \\ &= \frac{1}{8} \left(x - \frac{1}{4} \sin 4x \right) + C \\ &= \frac{x}{8} - \frac{1}{32} \sin 4x + C. \end{aligned}$$

$$(c) \quad (i) \quad V = \pi r^2 h$$

$$\therefore 27\pi = \pi r^2 h$$

$$r^2 h = 27$$

$$h = \frac{27}{r^2}$$

↓

Q13 (CONTD)

$$\text{new } A = 2\pi rh + \pi r^2$$

$$= 2\pi r \cdot \frac{27}{r} + \pi r^2$$

$$= \frac{54\pi}{r} + \pi r^2$$

$$= \pi \left(\frac{54}{r} + r^2 \right)$$

$$(11) \quad \frac{dA}{dr} = -54\pi r^{-2} + 2\pi r$$

$$\& \frac{d^2A}{dr^2} = 108\pi r^{-3} + 2\pi, > 0 \text{ for all } r.$$

$$\text{For max/min. let } \frac{dA}{dr} = 0$$

$$-\frac{54\pi}{r^2} + 2\pi r = 0$$

$$\frac{54\pi}{r^2} = 2\pi r$$

$$r^3 = 27$$

$$r = 3$$

Since $\frac{d^2A}{dr^2} > 0$ MIN at

$$\boxed{r=3}$$

Q14.

$$y = x^3 + px + q.$$

$$\frac{dy}{dx} = 3x^2 + p$$

now slope of L is 2

$$\therefore \text{ at } (1, 7) \quad m = 2$$

$$\therefore \begin{cases} 3 + p = 2 \\ p = -1 \end{cases} \quad 4$$

$$\text{And } 7 = 1^3 + -1 \times 1 + q.$$

$$\boxed{q = 7}$$

$$A = \int_0^1 [(x^3 - x + 7) - (2x + 5)] dx.$$

$$= \int_0^1 (x^3 - 3x + 2) dx$$

$$= \left[\frac{x^4}{4} - \frac{3x^2}{2} + 2x \right]_0^1$$

$$= \frac{1}{4} - \frac{3}{2} + 2.$$

$$= \left| \frac{3}{4} \right|$$

2.

Q14 (CONTD)

$$(1) \quad y = \frac{x-2}{\sqrt{x+1}}$$

$$\frac{dy}{dx} = \frac{\sqrt{x+1} \cdot 1 - (x-2) \times \frac{1}{2} (x+1)^{-\frac{1}{2}}}{x+1}$$

$$= \frac{\sqrt{x+1} - \frac{(x-2)}{2\sqrt{x+1}}}{x+1}$$

$$= \frac{2(x+1) - (x-2)}{2\sqrt{x+1}(x+1)}$$

$$y' = \frac{x+4}{2(x+1)\sqrt{x+1}}$$

$$(11) \quad y = \ln(\ln(2x+1))$$

$$y' = \frac{\frac{2}{2x+1}}{\ln(2x+1)}$$

$$y' = \frac{2}{(2x+1)\ln(2x+1)}$$

Q12 (a) now $\int_a^b \frac{f(x) - g(x)}{f(x) - g(x)} \cdot dx$

$$= \int_a^b 1 \cdot dx$$

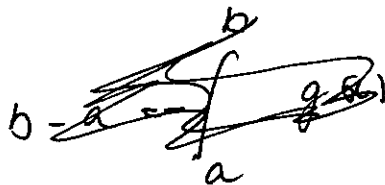
$$= [x]_a^b$$

$$= b - a.$$

Also $\int_a^b \frac{f(x) - g(x)}{f(x) - g(x)} dx = \int_a^b \frac{f(x)}{f(x) - g(x)} \cdot dx -$

$$\int_a^b \frac{g(x)}{f(x) - g(x)} \cdot dx$$

$$= 5 - \int_a^b \frac{g(x)}{f(x) - g(x)} \cdot dx$$

$\therefore$ 

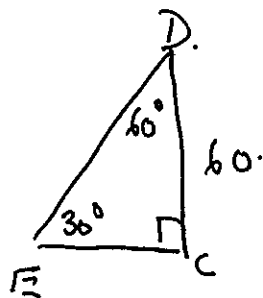
$\therefore b - a = 5 - \int_a^b \frac{g(x)}{f(x) - g(x)} \cdot dx$

$$\int_a^b \frac{f(x)}{f(x) - g(x)} dx = 5 - (b - a)$$

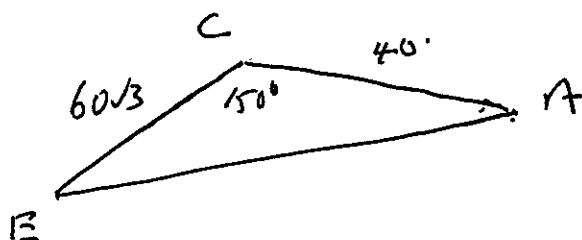
$$= \underline{5 + a - b}$$

Q15 (contd)

(b) (i)



$$\therefore EC = 60 \tan 60^\circ = 60\sqrt{3}$$



$$EA^2 = (60\sqrt{3})^2 + 40^2 - 2 \times 60\sqrt{3} \times 40 \times \cos 150^\circ$$

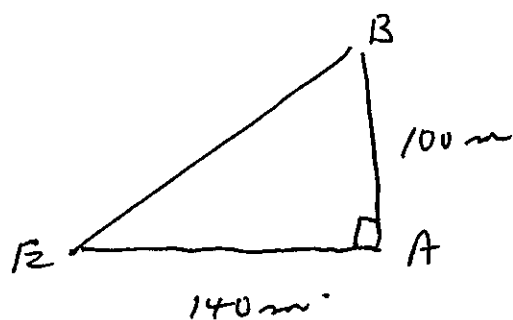
$$= 10800 + 1600 - 2 \times 60\sqrt{3} \times 40 \times -\frac{\sqrt{3}}{2}$$

$$= 12400 + 7200$$

$$= 19600$$

$$\therefore EA = 140 \text{ m.}$$

(i)



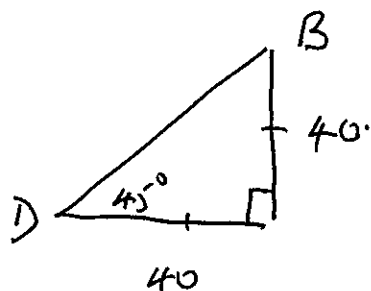
The angle of elevation of B from E.

$$\text{is } \angle BEA = \tan^{-1} \frac{100}{140}$$

$$= \tan^{-1} \frac{5}{7}$$

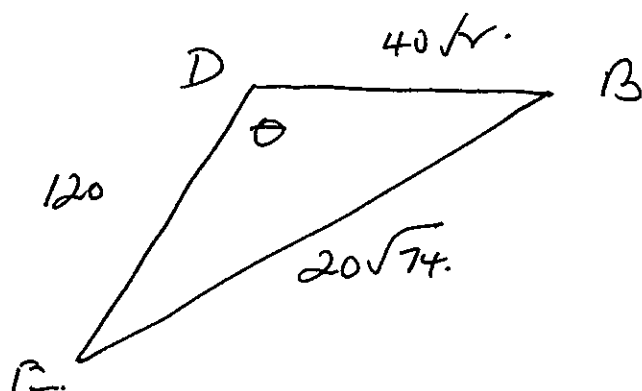
$$\approx 35^\circ 32'$$

Q15 (contd)



$$DB^2 = 40^2 + 40^2$$

$$\therefore DB = 40\sqrt{2}$$



$$EB^2 = 140^2 + 100^2$$

$$= 29600$$

$$\therefore EB = 20\sqrt{74}$$

$$\cos \angle EDB = \frac{120^2 + (40\sqrt{2})^2 - (20\sqrt{74})^2}{2 \times 120 \times 40\sqrt{2}}$$

$$= \frac{14400 + 3200 - 29600}{9600\sqrt{2}}$$

$$\cos \theta = -\frac{5\sqrt{2}}{2}$$

$$\text{now } \sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta + \frac{50}{64} = 1$$

$$\sin^2 \theta = 1 - \frac{25}{32}$$

$$= \frac{7}{32}$$

$$\sin \theta = \frac{\sqrt{7}}{4\sqrt{2}}$$

$$= \frac{\sqrt{14}}{8}$$

Q15 (CONTD)

(ii) $\therefore$ Area $y \triangleq EDB$

$$= \frac{1}{2} \times 120 \times 40\sqrt{2} \times \frac{\sqrt{14}}{8}$$

$$= \underline{600\sqrt{7}} \text{ m}^2 \quad \left(\text{Approx. } 1587.5 \text{ m}^2 \right)$$